DESCENT OF ADEQUACY ALONG A RING MAP WITH AN EXACT SEQUENCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file adequate.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The pushforward and base change lemmas making the comparison functors adequate. Same chapter, Lemmas lemma-pushforward-adequate and lemma-base-change-adequate, invoked at proof lines 1020-1022. Show the two functors appearing in the exact sequence are adequate.
- Kernels of maps of adequate functors are adequate. Same chapter, Lemma lemma-kernel-adequate, invoked at proof lines 1023-1024. Gives adequacy of F as the kernel.

Source reference. Original lemma 06V9: sources/stacks-adequate/pinned-original-source.tex, source lines 953–953. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06V3: sources/stacks-adequate/pinned-original-source.tex, source lines 709–709. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06VA: sources/stacks-adequate/pinned-original-source.tex, source lines 968–968. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $A \to A'$ be a ring map and let F be a module-valued functor on Alg_A such that*

(1) *the restriction F' of F to the category of A'-algebras is adequate, and*
(2) *for any A-algebra B the sequence*

$$0 \to F(B) \to F(B \otimes_A A') \to F(B \otimes_A A' \otimes_A A')$$

is exact.

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Then F is adequate.

Proof. The functors $B \to F(B \otimes_A A')$ and $B \mapsto F(B \otimes_A A' \otimes_A A')$ are adequate, see Lemmas 06VA and 06V9. Hence F as a kernel of a map of adequate functors is adequate, see Lemma 06V3. $\square$